Continuum Scaling of the Link Weight in a Four-Dimensional Lattice Path Sum

Sayed Heyder Ali Zia

1 Question and scope

We consider an infinite four-dimensional Euclidean hypercubic lattice with spacing Δ. We assign to each path γ from the origin to a lattice point x the weight

$$w(\gamma) = e^{-m_0 L(\gamma)}, \tag{1.1}$$

where we use $L(\gamma)$ for the number of links in the path.

We seek the dependence $m_0 = m_0(\Delta)$ that keeps long paths relevant as $\Delta \to 0$. We focus on this dependence rather than on evaluating the full propagator.

2 Assumptions and notation

We make the following assumptions:

(i) We work on an infinite Euclidean lattice.

(ii) We allow only nearest-neighbour steps.

(iii) We allow backtracking, repeated sites, repeated links, and self-intersections.

(iv) We count different ordered sequences of steps as different paths.

(v) We assign equal weights to all directions.

(vi) We sum over all finite path lengths, including the zero-link path at the origin.

(vii) We keep the physical endpoint separation fixed as we take the continuum limit.

We write the lattice sites as

$$x = \Delta n, \qquad n = (n_1, n_2, n_3, n_4) \in \mathbb{Z}^4. \tag{2.1}$$

We distinguish the physical position x from the dimensionless integer coordinate n.

More precisely, we take a sequence of lattice points $x_\Delta = \Delta n_\Delta$ approaching a fixed continuum point x. We suppress the additional subscript when it causes no ambiguity.

We denote the number of links by N and the physical length by ℓ:

$$N = L(\gamma), \qquad \ell = N\Delta. \tag{2.2}$$

Since N is dimensionless, we identify m_0 as a dimensionless cost per link. We reserve m for the continuum mass, which has dimensions of inverse length.

We use units $\hbar = c = 1$. We adopt the prescribed Euclidean weight as the physical input:

$$A(\gamma) = m_0 N, \qquad w(\gamma) = e^{-A(\gamma)}. \tag{2.3}$$

We take m_0 to be real and seek a stable continuum limit with $m^2 \geq 0$.

3 Path counting and the residual length weight

We denote the four coordinate unit vectors by e_μ, with $\mu = 1, 2, 3, 4$. In lattice units, we choose each step from

$$\{\pm e_1, \pm e_2, \pm e_3, \pm e_4\}. \tag{3.1}$$

We therefore have eight possible directions at every step.

If we do not restrict the endpoint, we count 8^N paths with N links. We obtain the total weight

$$8^N e^{-m_0 N} = e^{-(m_0 - \ln 8)N}. \tag{3.2}$$

We define

$$\delta(\Delta) = m_0(\Delta) - \ln 8, \qquad q(\Delta) = 8e^{-m_0(\Delta)} = e^{-\delta(\Delta)}. \tag{3.3}$$

We interpret δ as the residual cost per step after accounting for the exponential number of paths.

We now let $C_N(n)$ denote the number of N-step paths from the origin to the specified endpoint n. We write

$$\mathcal{G}_\Delta(\Delta n; m_0) = \sum_{N=0}^{\infty} C_N(n) e^{-m_0 N}. \tag{3.4}$$

We introduce

$$P_N(n) = \frac{C_N(n)}{8^N}. \tag{3.5}$$

We use $P_N(n)$ as a counting tool: we recognize it as the endpoint probability for an unbiased walk whose eight step directions each have probability $1/8$.

We obtain the exact rewriting

$$\boxed{\mathcal{G}_\Delta(\Delta n; m_0) = \sum_{N=0}^{\infty} q^N P_N(n).} \tag{3.6}$$

We have separated the endpoint constraint, contained in $P_N(n)$, from the additional suppression of long paths, contained in q^N.

4 Critical value and convergence

We identify the critical value

$$\boxed{m_{0,c} = \ln 8.} \tag{4.1}$$

For fixed lattice endpoint n, we use the large-N estimate

$$P_N(n) \sim c N^{-2}, \qquad c > 0, \tag{4.2}$$

along the allowed parity subsequence.

We understand the power N^{-2} from diffusion: we spread over distances of order $\sqrt{N}$ in four directions and therefore over a volume of order $(\sqrt{N})^4 = N^2$.

We must respect the parity restriction

$$N \equiv \sum_{\mu=1}^{4} n_\mu \pmod 2. \tag{4.3}$$

We obtain no paths for the opposite parity because each step changes the parity of the sum of the coordinates.

We distinguish three cases:

1. For $m_0 < \ln 8$, we have $q > 1$. We obtain exponential growth that overwhelms the power-law decay of $P_N(n)$, so the sum diverges.
2. For $m_0 > \ln 8$, we have $q < 1$. We obtain exponential suppression of long paths and a convergent sum.
3. For $m_0 = \ln 8$, we have $q = 1$. We still obtain a finite fixed-endpoint sum in four dimensions because $\sum_N N^{-2}$ converges.

We therefore retain $m_0 = \ln 8$ as an exception to the statement that every fixed value leads either to divergence or to the suppression of long-path continuum behaviour.

We also distinguish convergence at one endpoint from convergence after summing over every endpoint. At criticality, we have

$$\sum_n \mathcal{G}_\Delta(\Delta n; \ln 8) = \sum_{N=0}^{\infty} \sum_n P_N(n) = \sum_{N=0}^{\infty} 1 = \infty. \tag{4.4}$$

5 The meaning of long paths

We first consider a shortest path to n. We need

$$N_{\min} = \sum_{\mu=1}^{4} |n_\mu| \tag{5.1}$$

links.

At fixed nonzero physical separation, we have $N_{\min} \sim \Delta^{-1}$. We therefore cannot characterize the desired regime merely by saying that N becomes large: even a shortest path has infinitely many links in the continuum limit.

We instead seek fluctuating paths on the diffusive scale. We write their displacement after N unbiased steps as

$$X_N = \Delta \sum_{j=1}^{N} s_j. \tag{5.2}$$

We have

$$\langle s_j \rangle = 0, \qquad |s_j|^2 = 1, \qquad \langle s_j \cdot s_k \rangle = 0 \quad (j \neq k). \tag{5.3}$$

We consequently obtain

$$\langle |X_N|^2 \rangle = N\Delta^2. \tag{5.4}$$

We identify the scale on which a fluctuating walk explores a finite physical distance as

$$\boxed{N\Delta^2 = \text{finite}, \qquad N \sim \Delta^{-2}.} \tag{5.5}$$

We use this relation as a scaling statement, not as a claim that every contributing path has the same length.

We also find

$$\ell = N\Delta \sim \Delta^{-1}. \tag{5.6}$$

We therefore allow divergent microscopic path lengths while keeping the endpoints a finite physical distance apart.

For fixed $m_0 > \ln 8$, we can make the loss of finite-distance propagation explicit. Since $P_N(n) \leq 1$ and $q < 1$, we have

$$0 \leq \mathcal{G}_\Delta(\Delta n; m_0) \leq \sum_{N=N_{\min}}^{\infty} q^N = \frac{q^{N_{\min}}}{1-q}. \tag{5.7}$$

We obtain exponential decay as $\Delta \to 0$ at fixed nonzero separation.

We do not infer that exactly shortest paths alone contribute. We regard the shortest-path wording as a heuristic description, since longer paths can have substantial multiplicity.

6 Dependence of m_0 on Δ

We retain from (3.6) the residual length factor

$$q^N = e^{-\delta(\Delta)N}. \tag{6.1}$$

We consider the diffusive limit

$$N\Delta^2 \longrightarrow u, \qquad 0 < u < \infty. \tag{6.2}$$

We rewrite the exponent as

$$\delta(\Delta)N = \frac{\delta(\Delta)}{\Delta^2}(N\Delta^2). \tag{6.3}$$

We obtain a definite finite limiting suppression by requiring

$$\frac{\delta(\Delta)}{\Delta^2} \longrightarrow \alpha, \qquad 0 \leq \alpha < \infty. \tag{6.4}$$

We then find

$$q^N \longrightarrow e^{-\alpha u}. \tag{6.5}$$

Since $\delta(\Delta) = m_0(\Delta) - \ln 8$, we obtain

$$\boxed{m_0(\Delta) = \ln 8 + \alpha\Delta^2 + o(\Delta^2), \qquad \alpha \geq 0.} \tag{6.6}$$

6.1 Explanation of Equation (6.6)

We obtain (6.6) directly from the limit in (6.4). We can express that limit by writing

$$\frac{\delta(\Delta)}{\Delta^2} = \alpha + r(\Delta), \qquad r(\Delta) \longrightarrow 0.$$

We multiply by Δ^2:

$$\delta(\Delta) = \alpha\Delta^2 + \Delta^2 r(\Delta).$$

We denote $\Delta^2 r(\Delta)$ by $o(\Delta^2)$ because its ratio to Δ^2 vanishes. We then substitute $\delta = m_0 - \ln 8$ and obtain (6.6).

We can also understand the result physically. We have $N \sim \Delta^{-2}$ for the long fluctuating paths of interest. We therefore require the residual cost per link to be of order Δ^2, so that its product with the number of links remains finite.

We assign α dimensions of inverse length squared. We consequently keep $\alpha\Delta^2$ dimensionless, consistently with the dimensions of m_0.

We additionally require $m_0(\Delta) \geq \ln 8$ at sufficiently small finite spacing. We cannot approach the critical value from below while retaining the unrestricted positive path sum, because the sum would diverge at each such spacing.

6.2 Identification of the continuum mass

We identify the coefficient α using the one-step momentum dependence.

We let p denote physical Euclidean momentum and define

$$\varphi_\Delta(p) = \left\langle e^{ip\cdot\Delta s} \right\rangle = \frac{1}{8}\sum_{\mu=1}^{4} \left(e^{i\Delta p_\mu} + e^{-i\Delta p_\mu} \right). \tag{6.7}$$

We use $p \cdot x = \sum_{\mu=1}^{4} p_\mu x_\mu$ and $p^2 = \sum_{\mu=1}^{4} p_\mu^2$.

We expand at fixed p:

$$\varphi_\Delta(p) = \frac{1}{4}\sum_{\mu=1}^{4} \cos(\Delta p_\mu) = 1 - \frac{\Delta^2 p^2}{8} + O(\Delta^4 |p|^4). \tag{6.8}$$

We combine this factor with the residual length weight:

$$\log\left(q^N \varphi_\Delta(p)^N \right) = -N\delta(\Delta) + N \log \varphi_\Delta(p). \tag{6.9}$$

Using $N\Delta^2 \to u$ and $\delta(\Delta) = \alpha\Delta^2 + o(\Delta^2)$, we obtain

$$q^N \varphi_\Delta(p)^N \longrightarrow \exp\left[-u\left(\alpha + \frac{p^2}{8} \right) \right]. \tag{6.10}$$

We have discarded terms proportional to $N\Delta^4$, which vanish because $N\Delta^4 = (N\Delta^2)\Delta^2 \to 0$.

We define the proper-time parameter

$$s = \frac{u}{8} = \lim_{\Delta\to 0} \frac{N\Delta^2}{8}. \tag{6.11}$$

We then write the limiting factor as

$$e^{-s(p^2+8\alpha)}. \tag{6.12}$$

We identify the continuum mass through

$$m^2 = 8\alpha. \tag{6.13}$$

We therefore obtain

$$\boxed{m_0(\Delta) = \ln 8 + \frac{m^2\Delta^2}{8} + o(\Delta^2).} \tag{6.14}$$

We identify s as a parameter with dimensions of length squared. We distinguish it from both the link count N and the physical path length $N\Delta$.

We may choose

$$m_0(\Delta) = \ln(8 + m^2\Delta^2) \tag{6.15}$$

as one function satisfying (6.14). We do not require this particular finite-spacing choice; we require its asymptotic behaviour.

6.3 Limiting cases

We distinguish the following possibilities:

1. If $\delta(\Delta)/\Delta^2 \to +\infty$, we suppress paths with any fixed positive value of $N\Delta^2$. We lose the desired finite-mass diffusive limit.
2. If $\delta(\Delta)/\Delta^2 \to \alpha > 0$, we retain long paths with finite suppression. We obtain a finite positive mass squared $m^2 = 8\alpha$.
3. If $\delta(\Delta)/\Delta^2 \to 0$ with $\delta(\Delta) \geq 0$, we retain the diffusive regime without a residual mass suppression. We obtain the massless scaling limit.
4. If $\delta(\Delta) < 0$, we encounter a divergent unrestricted path sum at that spacing.

We therefore require both an approach to the critical value and control over the rate of that approach. We cannot determine the continuum behaviour from $m_0(\Delta) \to \ln 8$ alone.

7 Final result

We find the general finite-suppression scaling

$$\boxed{m_0(\Delta) - \ln 8 = \alpha\Delta^2 + o(\Delta^2), \qquad \alpha \geq 0,} \tag{7.1}$$

with $m_0(\Delta) \geq \ln 8$ at sufficiently small finite spacing.

For a specified continuum mass m, we write

$$\boxed{m_0(\Delta) = \ln 8 + \frac{m^2\Delta^2}{8} + o(\Delta^2).} \tag{7.2}$$

We interpret $\ln 8$ as the compensation for the exponential number of available walks. We interpret the correction of order Δ^2 as the residual suppression that remains finite for paths with $N \sim \Delta^{-2}$.

We retain $m_0 = \ln 8$ as the critical massless case. We also retain the distinction between the survival of long-path contributions and the separate normalization of a continuum propagator.

AI use declaration

Model(s) used: We used ChatGPT (Codex; the exact underlying model version was not available in the conversation).

Revisions/corrections made to model output, and clarifications added: We made many revisions and clarifications, including the distinction between link count and physical length, the dimensions of m_0, the fixed-separation continuum limit, the critical exception $m_0 = \ln 8$, the interpretation of long paths, the qualification of shortest-path domination, and the derivation of Equation (6.6).

Comparisons/checks against the lecture notes: We made two comparisons with the relativistic path-integral discussion in Chapter 6, printed pages 52–55: (1) the cancellation between path multiplicity and the exponential length weight; (2) the diffusive scaling in which the number of steps times the squared step length remains finite. We accounted for the different conventions for link count and physical length.

Calculation steps, limiting cases, or consistency checks verified by hand: We report one independent hand check.

Calculation steps, limiting cases, or consistency checks subjected to automated verification (via Mathematica, Python, LLMs): We used one LLM-assisted review of the scaling argument and its algebraic consistency. We did not perform an independent Mathematica or Python verification.